

Plexus

A blueprint for differentiable tissue

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Abstract

We introduce *Plexus*, a formal language for biological modelling built on a common abstraction of living systems. Plexus is constructed by decomposing the large existing corpus of simulation models into a common representation. This includes neural computation (Hodgkin–Huxley, connectome and neural-network models), multicellular dynamics (vertex, Cellular Potts and active-matter models), transport processes (diffusion, chemotaxis and reaction–diffusion), and developmental dynamics (embryogenesis and morphogenesis).

Within Plexus, every simulation model is expressed through five primitives: *sets* of biological entities, the *states* carried by each entity, continuous spatial *fields*, *maps* describing relationships between entities and representations, and *operators* implementing biological mechanisms. Operators are further organized by the hierarchy of interactions they implement (set–set, set–field, field–field, and parent–child). Every primitive is paired with a differentiable counterpart: operators become differentiable functions, maps become differentiable transformations, and fields may be represented explicitly or as neural implicit representations. Models originating from different simulation paradigms can therefore be expressed, composed, compared, and optimized within a single formalism without sacrificing mechanistic interpretability.

This representation enables mechanistic analysis that goes beyond forward modelisation. Competing models can be fitted against biological observations, their residuals compared, and persistent discrepancies interpreted as evidence for missing mechanisms rather than suboptimal parameters. Plexus provides a common language in which simulation, inverse modelling, mechanistic analysis, and hypothesis generation become successive stages of mechanistic model discovery.

Part I — The abstraction

1 Two entities: sets and fields

The container is built from two kinds of object: **sets** and **fields**. A **set** S_k is a discrete collection of like elements at one scale — particles, cells, populations, organisms, or the compartments inside a cell: membrane particles, cytoplasm, nuclei, and metabolites (Fig. 1). Every element of a set shares the same state schema and the same operators, and a set is stored as batched tensors — never one object per element.

Sets are linked by **containment**. A parent map

$$\pi : S_k \longrightarrow S_{k+1}$$

sends each element to the parent that contains it — a particle to its cell, a cell to its population. A cell is therefore not a single object but the collection of everything that maps to it: its particles, nucleus, metabolites, and any other child sets. Stacked across scales, these maps form the *containment graph* of the next section.

A **field** $f : \Omega \rightarrow \mathbb{R}^c$ is a continuous quantity defined on its own grid — morphogen concentration, pressure, substrate stiffness, or an extracellular chemical. Fields are *not* part of the containment graph: they do not contain cells, and cells do not contain fields. Instead, each field is coupled to one or more sets through *exchange* operators — secretion, sensing, particle-grid transfer, or field sampling (Section 4).

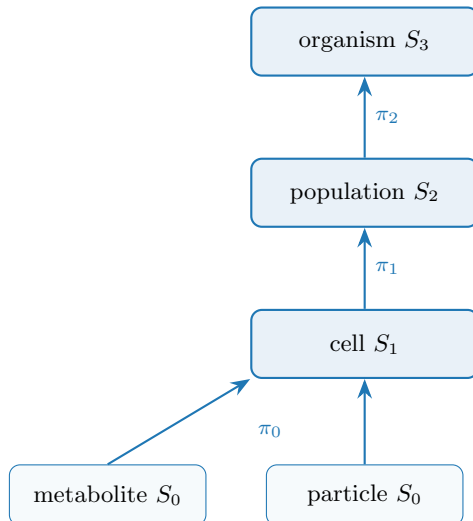


Figure 1: Containment across scales: each parent map π sends an element to the one that contains it. A cell is the collection of everything that maps to it — here its particles and metabolites; in general also its membrane, nucleus, and other child sets (Section 3).

2 State: what each element carries

A set says *what exists*; its **state** says *what each element carries*. State is a first-class concept, not an implementation detail: every set declares a typed record of named quantities — its *state schema* — and that record is the sole interface between the set and every operator, plotter, and integrator that touches it. A particle carries **pos**, **vel**, and the MPM continuum blocks **F**, **C**; a neuron carries **voltage**, **calcium**, **adaptation**; a cell carries **polarity**, **stiffness**, **growth**; a synapse carries its gating and plastic weight. A field carries state too — its grid values — but the per-element record is what makes the sets interchangeable.

set	state (named blocks)	operators that transform it
particle	pos, vel, F, C	MPM strain/scatter/gather, boids
neuron	voltage, calcium, adaptation	neuron_voltage, signal
cell	polarity, stiffness, growth	division, adhesion, polarity align
synapse	g (gating), plastic_w	synapse_ode

Two consequences make state the pivot of the whole language.

Operators are state $\rightarrow$ state. An operator reads some named blocks of a set’s state and returns a derivative on others — nothing else. `neuron_voltage` reads `voltage` and the aggregated synaptic current and returns $d\text{voltage}$; `mpm_strain` reads `F` and returns its update; a boid rule reads `pos/vel` and returns an acceleration. Because the engine integrates whatever block the schema declares dynamical (Section 5), a new kind of dynamics is a new state block plus an operator that returns its derivative — never an engine change. This is why `pos/vel` is not privileged: it is simply the state a moving particle happens to carry.

The frameworks become specifications. Once state is first-class, the sibling frameworks stop being separate code and become different *states* over the same five concepts. `connectome-cx` is the specification with sets `network`, `neuron`, `synapse` and operators `neuron_voltage`, `synapse_ode`; an embryogenesis model reuses the identical set machinery with operators `cell_grow`, `active_stress`, and a cardiac model swaps in `active_force` — same language, different vocabulary. The engine, the maps, and the schedule are identical; only the state schemas and the operator list change. The whole framework is then five concepts — **sets, state, fields, maps, operators** — and everything else is a specification written in them.

Anatomy is structure; assemblies are emergent. Structure stays anatomical: the neural hierarchy is `network` $\rightarrow$ `region` $\rightarrow$ `column` $\rightarrow$ `neuron` $\rightarrow$ `compartment`, each a set contained in the next. A functional *assembly* is deliberately *not* a level in this tree — it is an emergent subset selected by a map or a state (`neuron[active=1]`), computed on demand — so anatomy and computation are never conflated inside the containment graph.

Operators are contracts; implementations are backends. A state $\rightarrow$ state operator is a *contract*, not a piece of biophysics. `neuron_voltage` is the contract “membrane state $\rightarrow$ its derivative”; its implementations — Hodgkin–Huxley, Izhikevich, Morris–Lecar, FitzHugh–Nagumo, a Neural ODE, an MLP, a Transformer — are interchangeable, and the spec selects one without touching the maps, the schedule, or the rest of the model (`{op: neuron_voltage, implementation: hodgkin_huxley}`). Likewise `synapse_ode` is a contract with conductance, alpha, plastic, and complex-phase implementations. The two most developed neural frameworks are then two *backends* of the *same* contracts, not two engines: **Jaxley** implements `neuron_voltage` as multicompartment Hodgkin–Huxley and `synapse_ode` as a conductance synapse, differentiating through an implicit solver; a connectome **GNN** implements the same contracts as learned message functions, promoting the scalar voltage to a coupled (v, c) state and rotating the postsynaptic response by a broadcast neuromodulator α .

Morphology is Sets, not an embedding. The two differ only in *resolution*, and even that is structural. Plexus keeps morphology as multiscale sets — `neuron` $\rightarrow$ `compartment` $\rightarrow$ `ion_channel` — so Jaxley’s explicit compartments are the fine-grained end of the same containment hierarchy and its channels are operators on the `compartment` set. The GNN’s geometry embedding s_i is a *model reduction* of that hierarchy: one implementation that summarises the compartment tree in a frozen per-neuron feature. Explicit compartments and a geometry embedding are thus two *resolutions* of one hierarchy, not two data models — which is what makes mechanism search well posed: holding the sets, states, and maps fixed, the search ranges over the implementation of each contract and the resolution of each hierarchy.

An intermediate representation, not a simulator. Taken together this is the thesis of Plexus. It is not another simulator but a typed *intermediate representation* for biological computation: a model expressed in NEURON, Jaxley, Flyvis, connectome-cx, an embryogenesis solver, or a cardiac model compiles to the same sets, states, maps, fields, operators, and schedule, and each existing simulator becomes a *backend* implementing one family of operator contracts. Plexus is to a bio-physics simulator what LLVM is to a C compiler — the payoff is portability (one model, many backends) and mechanism search, not a faster solver.

3 Maps between sets

Section 1 named the sets; this section connects them. Sets are connected by named **maps**, and there is only one primitive here: a map $m : S_a \rightarrow S_b$ is an index relation sending each element of one set to an element of another. Hierarchy, connectivity, signalling, transport, and compartmental organization are all expressed through collections of such maps; the engine manipulates maps uniformly, and only their biological interpretation differs. This is the third primitive, on the same footing as sets and fields, and it is what lets the framework grow without special cases: the container is not (S, F, π) — sets, fields, and a single hardcoded parent map — but (S, F, M) , sets and fields over an *open collection of named maps*. An operator names the map it traverses (**via: parent**, **via: post**) and never branches on which.

We name a map by the role it plays — a convention for the reader, not a distinction the engine makes. A **parent map** $\pi : S_k \rightarrow S_{k+1}$ sends a child to the one parent that owns it; stacked across scales the parent maps form the hierarchy (an acyclic tree), and biologists call this relation *containment*. A **pre** or **post map** $\iota : S_e \rightarrow S_k$ sends an *edge-entity* (a synapse, a junction, a vessel segment) to one of its endpoints, so a stateful connection carries several such maps at once; biologists call this *incidence*. The parent map builds the tree; pre/post maps overlay a graph on it. We take the parent map first, since it defines the hierarchy, then pre/post maps, which let recurrent circuits and other stateful connections live inside that hierarchy without breaking the tree.

The parent map: the hierarchy

Two points make the parent map precise, and they are what keep the engine free of branching.

One set per kind of object. Rather than a single set whose elements carry a label saying what each one is — which forces every operator to ask “what am I looking at?” at run time — we keep *one set per kind*: membrane particles, nuclei, metabolites, and cells are separate sets, each with its own state schema, operators, and fields. An operator then names the one set it acts on (**at: nucleus**) and never branches. This is the same discipline an entity–component system or a relational database already uses.

Joined by parent maps. The sets are joined by parent maps, declared simply by naming the parent,

$$\pi : S_k \longrightarrow S_{k+1},$$

which send every element to the parent that contains it. A parent may have many children, so the parent map is a **graph**, not a chain: one cell gathers several child sets at once (Fig. 2). The engine needs only two things — **Level.parent** (which parent each element has) and **Level.parent_name** (which set that parent belongs to) — so new biological structure is just a new set and a parent map, never an engine change.

A new set, or a new type? The rule (Table 1): two groups that differ in *state or operators* are *different sets*; two groups that share both and differ only in *parameter values* are *types within one set* — the **types:** block. A cell and a nucleus have different state and dynamics, so they are different sets; a soft cell and a stiff cell differ only in stiffness, so they are two types of the single cell set.

	Different sets	Types within one set
example	membrane vs. nucleus vs. metabolite	soft cell vs. stiff cell
differ in	state space + operators	only parameter values
in the spec	separate sets: with parent:	a types: block inside one set
dispatch	by name (at: nucleus)	per-element parameter lookup

Table 1: When to declare a new set versus a new type: a different *kind* (state space or dynamics) $\Rightarrow$ a new set; a parametric *variant* $\Rightarrow$ a type within a set.

Keeping each operator attached to its own set is the payoff: **membrane_tension** acts on **membrane_particle**, **metabolism** on **metabolite**, and **gene_regulation** on **nucleus**, while Exchange operators connect compartments explicitly (nucleus $\leftrightarrow$ cytoplasm, membrane $\leftrightarrow$ extracellular field) — never one mega-operator branching on a role flag.

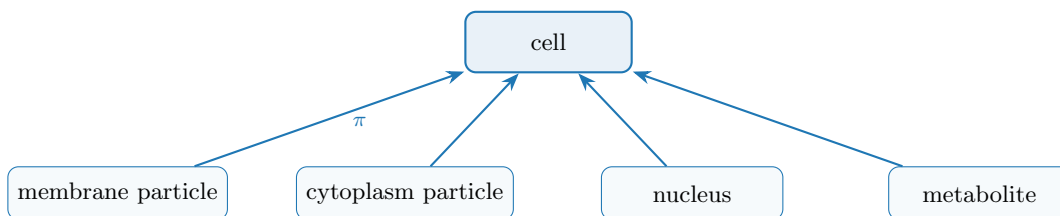


Figure 2: A containment graph: one parent set (**cell**) with several child sets, each with its own parent map. The cell is the collection of these children — not a bag of one kind of particle. A new compartment is a new row in the spec (Listing 1), not an engine edit.

Listing 1: A cell as a collection of child sets: each compartment is its own set with a parent map.

```

sets:
  cell: {n: 100}
  membrane_particle: {parent: cell, per_parent: 64, role: membrane}
  cytoplasm_particle: {parent: cell, per_parent: 200, role: cytoplasm}
  nucleus: {parent: cell, per_parent: 1, role: nucleus}
  metabolite: {parent: cell, per_parent: 50, role: metabolite}

```

Pre and post maps: stateful connections as edge-sets

The parent map is a tree: every element has exactly one parent. But a *connection* joins two elements and belongs to neither exclusively — a chemical synapse has one presynaptic and one postsynaptic neuron, and is owned by both and by neither. A connection is therefore not a parent map; it is another map (indeed a pair) on the same sets.

When a relation is enough, and when it is not. A connection has two faithful representations, and which one is correct is set by whether the connection has *state of its own*.

- **Passive.** If the edge is a fixed scalar — a synaptic weight W_{ij} , a spring constant — it is a **relation** $E \subseteq S_k \times S_k$ with an edge attribute: exactly the `Level.edge_index` of the previous section, read in place by a Lateral operator (Section 4).
- **Dynamical.** If the edge carries its own evolving state — synaptic gating and transmitter, short-term facilitation, a plastic weight, a remodelling adhesion — then it is promoted to an **edge-set** S_e : a first-class set whose *elements are the connections*, with their own state schema and their own operators, joined to the endpoint sets by **pre and post maps**.

This is the rule: *a relation suffices while an edge is passive; when an edge carries dynamical state it becomes an edge-set with its own pre/post maps*. The edge-set is not a new layer or a new engine primitive — it is an ordinary set that happens to sit on the connections, and it still has a parent (the **network**).

Pre and post maps. An edge-set element points at its endpoints through one map per endpoint role. For a directed synapse,

$$\iota_{\text{pre}} : S_{\text{synapse}} \rightarrow S_{\text{neuron}}, \quad \iota_{\text{post}} : S_{\text{synapse}} \rightarrow S_{\text{neuron}},$$

send each synapse to its presynaptic and postsynaptic neuron. They do not answer “who owns me” (the synapse’s parent is the **network**), they answer “whom do I connect.” Structurally they are index buffers on the edge-set (`.pre`, `.post`), the exact analogue of `.parent` — so an operator gathers or scatters along a pre/post map with the same machinery it uses along the parent map, and the engine gains no special case. The same shape recurs wherever a stateful connection does: ligand–receptor contacts ($\iota : S_{\text{ligand}} \rightarrow S_{\text{receptor}}$), cell–cell junctions ($\iota_1, \iota_2 : S_{\text{junction}} \rightarrow S_{\text{cell}}$), and vascular segments ($\iota_{\text{from}}, \iota_{\text{to}} : S_{\text{seg}} \rightarrow S_{\text{node}}$).

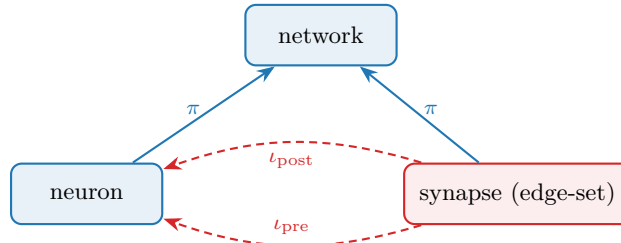


Figure 3: The neural hierarchy. Solid *parent* maps π (blue) make both **neuron** and **synapse** children of the one **network** — the tree. Dashed *pre/post* maps $\iota_{\text{pre}}, \iota_{\text{post}}$ (red) send each synapse edge-entity to the two neurons it connects — the graph. The synapse is a set like any other; it merely carries pre/post maps in addition to a parent.

Signalling reads off the maps. Endpoint voltages are *gathered* onto each synapse along its pre and post maps; the synapse integrates its own state; its output current, scaled by the synaptic weight W_e , is *aggregated* back onto the postsynaptic neuron; and the neuron integrates its voltage.

For synapse e and neuron i , writing $\text{pre}(e), \text{post}(e)$ for the maps,

$$\begin{aligned} \text{synapse (edge-set ODE): } \quad & \frac{ds_e}{dt} = h(s_e, v_{\text{pre}(e)}, v_{\text{post}(e)}; \theta_{\tau(e)}), \\ \text{synaptic current } (\uparrow \text{ post}) : \quad & I_i = \sum_{e: \text{post}(e)=i} W_e g(s_e, v_i), \\ \text{neuron (set ODE): } \quad & \tau_i \frac{dv_i}{dt} = \text{intrinsic}_i(v_i) + I_i + I_i^{\text{ext}}. \end{aligned}$$

The middle line is an **Aggregate** (Section 4) run along the **post** map rather than the **parent** map: the sum over children of a parent becomes a sum over the edges into i , each weighted by its synaptic weight W_e . A passive connectome is the degenerate case — drop the synapse state s_e , so g collapses to $\phi(v_{\text{pre}(e)})$ and $I_i = \sum_{e: \text{post}(e)=i} W_e \phi(v_{\text{pre}(e)}) = \sum_j W_{ij} \phi(v_j)$, recovering a single Lateral **signal** operator on a plain relation (no edge-set needed). Listing 2 is the stateful version.

Listing 2: A recurrent circuit as one hierarchy: a **neuron** set and a **synapse** edge-set, both contained in the **network**, the synapse joined to the neurons by **pre/post** incidence maps.

```
sets:
  network: {n: 1}
  neuron:                                     # an ordinary set
    parent: network
    per_parent: 13731
    types: {T4a: {sign: +1, tau: 0.02}, M1l: {sign: -1, tau: 0.05}} # Dale sign, tau
  synapse:                                   # an edge-set: elements ARE the connections
    parent: network                           # containment: owned by the network
    edge_set: true
    pre: neuron                               # incidence iota_pre
    post: neuron                              # incidence iota_post
    state: {g: 0.0, plastic_w: 0.0}           # its own dynamical state
operators:
  - {op: synapse_ode,      at: synapse}       # ds_e/dt = h(s_e, v_pre, v_post)
  - {op: synaptic_current, at: synapse, to: neuron} # aggregate output onto post neuron
  - {op: neuron_voltage,  at: neuron}         # tau dv/dt = intrinsic + I_syn + I_ext
schedule: [synapse_ode, synaptic_current, neuron_voltage]
```

4 Operators

A GNN is an **operator** $\mathcal{O}$: a map on **state** (Section 2) dispatched by the relation it acts on — it reads named blocks of a set’s state and returns a derivative on others. Equivalently, an operator changes exactly one of the things the container holds (Fig. 4): the **state** on the nodes, the **field** itself, the **relations** between them, or the **entities** themselves. **Four dynamics operators** move *state* along a fixed structure, one per relation the container admits, each returning a time-derivative contribution. **Field** operators are a field’s own dynamics — diffusion, decay, reaction — mapping field to field. **Rewire** changes the *relations* — the edge set E , which nodes interact. **Structural** operators change the *entities* — the node set $|S_k|$, which nodes exist: **Divide** and **Die** are its two members. **Seed** writes the initial state once, at $t = 0$. All but the first four return no state delta. Eight kinds in all, and the implementation (`plexus.models.base.KINDS`) recognises exactly these. The four dynamics operators:

Lateral	$\mathcal{O}_E, E \subseteq S_k \times S_k$	(ODE interaction + graph Laplacian)
Aggregate	$\uparrow \sum_m : S_a \rightarrow S_b$	(reduction along a map m)
Broadcast	$\downarrow m^* : S_b \rightarrow S_a$	(lift along a map m)
Exchange	$S, \mathcal{G} : S_k \leftrightarrow F$	(a map to/from a field)

Two pieces of notation: $S_k \times S_k$ is the set of all node *pairs*, so an edge set $E \subseteq S_k \times S_k$ is simply a graph on S_k — which nodes interact (e.g. neighbours within a cutoff radius, or a fixed connectome). And $\mathcal{S}, \mathcal{G}$ are the two halves of a field coupling: *scatter* $\mathcal{S} : S_k \rightarrow F$ writes object state into the field, while *gather* $\mathcal{G} : F \rightarrow S_k$ reads it back. Each operator in turn:

Lateral $\mathcal{O}_E$. Message passing *within* one scale along the edges E . It carries the within-scale physics: pairwise interactions (forces, adhesion, signalling) plus a graph Laplacian that diffuses or smooths quantities over the edges. Examples: particle–particle elasticity, cell–cell flocking or adhesion, neuron–neuron signal propagation.

Aggregate $\sum_m$ (up). Reduces along a map m : pools the elements that m sends to each target into one quantity. Along the **parent** map a cell’s position is the mean of its particles, its metabolic state the sum over its molecules; along the **post** map the current into a neuron is the sum over its incoming synapses (the middle equation of Section 3). Same operator, different map named by **via**..

Broadcast m^* (down). The adjoint of Aggregate: it lifts a target’s state back onto every element the map m sends to it. Along the **parent** map a decision taken at the cell level (a body force, a polarity, a signalling output) is applied identically to all of that cell’s particles; along the **pre/post** maps it gathers the endpoint voltages a synapse needs. Aggregate and Broadcast are a matched up/down pair on *any* named map, so no new operator is needed for a stateful connection: **synapse.ode** (a Lateral operator on the synapse edge-set) sits between a Broadcast that gathers $v_{\text{pre}}, v_{\text{post}}$ and an Aggregate that scatters the current onto the post neuron, and a passive connectome collapses the whole thing to a single Lateral **signal** operator on a plain relation.

Exchange $\mathcal{S}, \mathcal{G}$. A map whose other endpoint is a *field* rather than a set: scatter $\mathcal{S}$ deposits/secretes object state into the field, gather $\mathcal{G}$ samples/senses the field back onto the objects. It is exactly the MPM particle–grid transfer (P2G/G2P) and chemotactic secretion/sensing — Aggregate/Broadcast read as “set to field / field to set,” which is why Exchange is not a separate idea so much as the field-valued corner of the same map primitive.

Rewire $\mathcal{R}$ (relations). Rewire changes the *relation* $E \subseteq S_k \times S_k$ on which the lateral operators act — *which* nodes are neighbours — and returns no state delta. A *geometric* relation is rebuilt every tick from the live configuration (a **radius graph**: all pairs within a cutoff; or a k -nearest-neighbour or Delaunay graph), so the interaction graph tracks motion, growth, and division; a *fixed* relation (a connectome) is built once and never rewired. Rewire is of a different nature from the other five: it neither moves state nor changes who exists — it maintains the *scaffold* the dynamics run on. Separating *who* interacts (Rewire) from *how* they interact (Lateral) is also what turns a dense $O(N^2)$ force law into $O(|E|)$ message passing and lets several lateral operators share one graph.

Structural: Divide and Die (entities). The dynamics operators move *state* and Rewire moves *relations*, both while the node set stays fixed; living matter also rewrites its own membership. **Divide** $\mathcal{B} : x \mapsto \{x', x''\}$ (proliferation) and **Die** $\mathcal{D} : x \mapsto \emptyset$ (apoptosis / efflux) are the only operators that change the *entities* themselves — the cardinality $|S_k|$. For a *living* system they are as primordial as the four dynamics operators: a tissue is defined as much by the cells it makes and loses as by how they move. Everything else assumes the cells it acts on already exist.

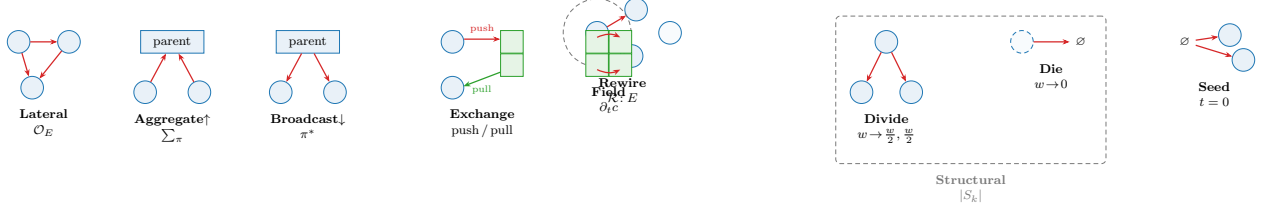


Figure 4: The eight operator kinds, grouped by what they change. The first four are *dynamics*: they move a set’s *state* along a fixed structure — *Lateral* within a set; *Aggregate/Broadcast* along a map (the parent map π , or a **pre/post** map); *Exchange* to/from a field (P2G/G2P and the metabolite–reaction graph are both Exchange). *Field* is a field’s own dynamics — diffusion, decay, reaction — field to field, returning no set delta. The rest return no state delta either: *Rewire* changes the *relations* — the edge set E , which nodes interact (a radius graph rebuilt each tick, dashed cutoff); *Structural* changes the *entities* — the node set $|S_k|$, and its two operators are *Divide* and *Die* (dashed box), cell division and death on a fixed buffer with per-element occupancy w ; *Seed* writes the initial state once, at $t = 0$, and never again.

Differentiating division and death. Changing $|S_k|$ is the one move that resists automatic differentiation: it alters a tensor’s shape, and “does this cell divide?” is a discrete event with no useful gradient. We keep the framework differentiable with a **continuous-occupancy** formulation. Rather than resize, we hold a fixed buffer of $N_{\max}$ slots and give every element a *mass* $w_i \in [0, 1]$: **Die** decays $w_i \rightarrow 0$ (the slot goes dormant, not deleted); **Divide** activates a dormant slot, copies the mother’s state with a small perturbation, and splits the mass $w \rightarrow (\frac{w}{2}, \frac{w}{2})$. *Every operator then honours w* : Aggregate becomes a *mass-weighted* reduction $(\sum_i w_i x_i) / \sum_i w_i$, Lateral weights each message by w_j , and so on — so the boolean selector mask of Section 7 is just the $\{0, 1\}$ special case of w . The trajectory between events is now smooth in w , so gradients flow on a fixed-shape tensor (and `torch.compile` still applies). The one genuinely non-differentiable part — the integer decision of *when* to divide or die — is handled separately: a straight-through or Gumbel estimator to learn a division *policy*, a score-function (REINFORCE) gradient for an expected objective with respect to division/death *rates*, or black-box search (UCB/CEM) when the objective is itself discrete (final count, fraction escaped). This is the *use both* recipe of Part III applied to structure: gradients for the continuous interior, black-box for the discrete choices.

5 Dynamics: a schedule

A simulation is a multi-rate composition of operators, declared as a **Schedule**. In math terms, one tick is the composition $\Phi = \mathcal{O}_n \circ \dots \circ \mathcal{O}_1$, and the dynamics is its iteration $x_{t+1} = \Phi(x_t)$ — a discrete flow. *Multi-rate* means the operators need not all fire on every tick: a slow process (diffusion, cell division) can be scheduled to act once every k ticks, while a fast one (mechanics) runs every tick, so a single **Schedule** interleaves dynamics that evolve on different time scales. The **Schedule** is just the code-level encoding of that composition. Each operator is *pure*: instead of editing the state in place, it only *returns its contribution* — a delta Δ_i (a time-derivative term) for the sets it touches. The schedule sums the deltas acting on each set and integrates once per tick.

An operator declares the *order* of its delta, since a time-derivative can be of position or of velocity. A *first-order* (overdamped) operator returns a velocity and integrates as $x_{t+1} = x_t + \Delta t \sum_i \Delta_i$; a *second-order* (inertial) operator returns an acceleration and integrates as $v_{t+1} = v_t + \Delta t \sum_i \Delta_i$, $x_{t+1} = x_t + \Delta t v_{t+1}$. The order is a fixed property of each operator (EMIT: **velocity**

for first order, **acceleration** for second): attraction–repulsion is first order, boids flocking is second; all engine-integrated operators acting on one set must agree, so the set integrates as a single order. (A dynamics operator may instead declare `EMIT = mpm_acceleration` — an acceleration consumed by the MPM substep as the external force a_{ext} rather than integrated by the engine.) Friction in the inertial case is itself an operator (a `drag` delta $-\kappa v$), not a special term in the integrator.

The integrated quantity is the set’s declared state. Position and velocity are only the *spatial* instance of this rule. What the engine integrates is whatever block a set’s *state schema* declares dynamical (Section 7): a moving particle declares `pos/vel`, but a neuron declares a scalar voltage, a metabolite a concentration, a gene its expression level, a synapse its gating and plastic weight. Each integrates under the same first-order rule $x_{t+1} = x_t + \Delta t \sum_i \Delta_i$ with x the declared state, and only the spatial blocks are subject to the world boundary. Making integration *schema-driven* rather than pos/vel-specific is the single change that lets one engine carry mechanics, signalling, metabolism, and gene expression at once — the temporal counterpart of generalizing a lone hardcoded parent map to the open map collection M . Voltage, calcium, receptor occupancy, and a plastic weight are then all first-class integrands, each just a schema block plus an operator that returns its derivative.

Two consequences follow: several operators may act on the same set in one tick (their deltas simply add — none overwrites another), and because nothing is mutated in place, gradients flow through the sums and the integrator, so the whole rollout stays differentiable. The LLM searches over schedules and operator parameterizations — one well-defined action space instead of per-domain code.

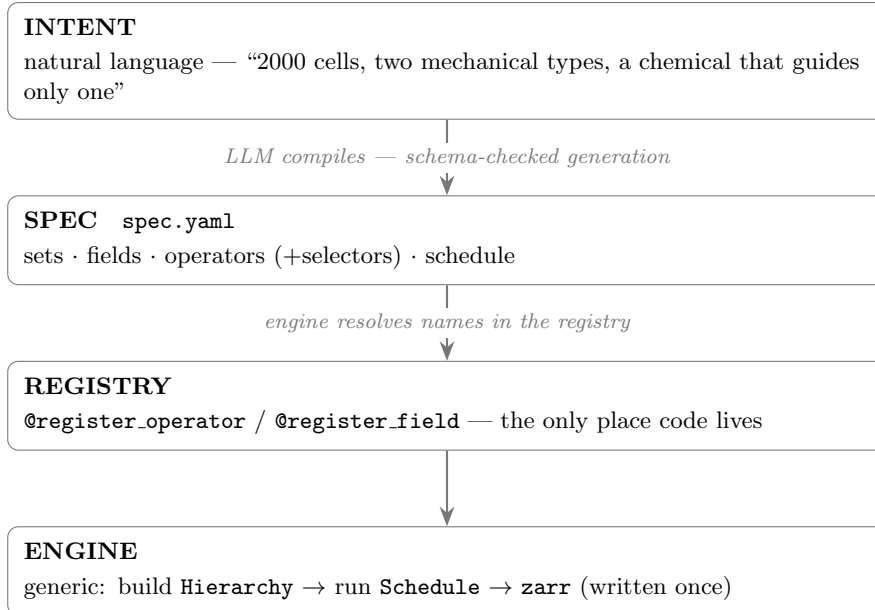
Part II — A formal intermediate representation

6 Two layers and the contract between them

The contribution of this part is not that a language model can drive the framework — it is that there is a **formal intermediate representation** for anything to drive at all. Between a person’s *intent* and a running *simulation* we interpose one typed, declarative artifact, the **spec**, and the system becomes a compiler stack:

$$\text{Intent} \longrightarrow \text{Language} \longrightarrow \text{Calculus} \longrightarrow \text{Simulation}.$$

Intent compiles to the **language** (the spec, `spec.yaml`); the spec names primitives in a **calculus** of operators (the registry); the engine **interprets** that calculus into a simulation. In one line: the spec is the *source*, the registered operators are the *primitives*, and the engine is the *interpreter*. A language model is one front end on the first arrow — useful, but replaceable; the durable object is the representation in the middle. Every property claimed below is a property of *it*, not of the model that writes it.



The **spec is the contract**. The LLM owns everything above it (intent → spec); the code owns everything below (operators). They never touch directly. This makes the system both LLM-drivable and human-auditable: the spec is small, typed, and reviewable, unlike the 500-line flat `config.py` it replaces.

Being a small, typed, declarative file gives the contract the properties one wants of an interface between an unreliable generator and exacting code:

- **Verifiable.** The spec is validated against the schema *before* anything runs: an unregistered operator, a missing required property, or a malformed selector is rejected up front, so a spec that loads is guaranteed runnable (the missing-operator and missing-property cases below). Beyond well-formedness, each simulation also ships an *intent* check that measures whether the requested behaviour actually occurred — “it ran” is not “it worked”.
- **Reproducible.** A run is bit-deterministic given (spec, seed) — the engine forces deterministic kernels — so the same spec yields the same trajectory, and a fresh seed draws a new realization of the same dynamics.
- **Saveable.** The spec is ~20 lines, so it version-controls, diffs, and archives trivially; together with the seed it *is* the complete record. We archive the recipe, not the dish: store spec + seed (plus metrics and a thumbnail) and regenerate the trajectory on demand (`archive.py`, `GALLERY.md`).
- **Composable.** A variant is a base spec plus a handful of overrides, which makes ablations and the optimizer’s parameter sweeps cheap.
- **Extensible.** New behaviour is one new registered operator; the operator vocabulary grows monotonically, and every earlier spec keeps working.

7 The simulation language

The contract above is, in effect, a small **domain-specific language** in declarative-data form. Like any language it has *words* (vocabulary), *grammar* (how they combine — here a typed schema, which *is* the language definition: required keys, type fractions summing to one, every operator registered, every schedule token resolving), and *semantics* (what running them means: the engine builds the **Hierarchy** and iterates the schedule, $x_{t+1} = \Phi(x_t)$). Table 2 is the complete vocabulary; every construct in it appears in the example of Listing 3.

Structurally this is an **entity–component–system** (ECS) architecture: sets are entities, their state buffers are components, operators are systems, and the schedule is the ECS system pipeline. We adopt that framing deliberately — it is the cleanest mental model and matches the engine one-to-one. Each entity kind declares its component layout once in the registry (`@register_entity`): a typed *state schema* naming which state columns are position, velocity, and so on, plus *render* hints (what to colour by, whether to draw orientation). The engine, the operators, and the visualizer all read column semantics from this one declaration rather than hard-coding offsets like `state[:, :2]` — so a new entity is drawn and integrated correctly for free.

Most of the spec is plain typed key–value; the only genuine sub-grammars are the last two blocks of Table 2 — *selectors* and the *schedule*. Both are deliberately minimal but extend naturally: selectors to conjunctions and to geometric or temporal predicates (`cell[type=soft & loaded=0]`, `cell[x<0.5]`, `cell[age>10]`), and the schedule to explicit per-operator rates and nested loops.

We keep the language as *declarative data* (YAML/JSON) rather than a bespoke syntax, so an LLM can emit it under schema-constrained decoding and a human can diff, review, and archive it; the grammar is pinned by a typed schema (Pydantic or JSON-Schema), and composition, overrides, and parameter sweeps (variants, ablations, the optimizer) are delegated to a config layer such as Hydra rather than reinvented. Prior art worth borrowing: ECS engines (the scheduling model), SBML/Antimony (reaction–species vocabulary), NeuroML and MorpheusML/PhysiCell (multicellular specs), UFL (PDE operators), and NetLogo/Mesa (agent rules).

A complete spec is a single `spec.yaml`. Listing 3 assembles the vocabulary of Table 2 into one: two cell types, particles contained in each cell, one chemical field, and a schedule. Soft cells deposit the chemical and chemotax (climb) it, so they self-aggregate, while stiff cells — selected out by `cell[type=soft]` — do not.

Listing 3: A complete simulation: soft cells secrete and follow a chemical and self-aggregate.

```
general: {name: aggregate, seed: 0, n_frames: 250, dt: 0.05, boundary: periodic}
sets:
  cell:
    n: 100
    types: {soft: {fraction: 0.5, youngs: 12}, stiff: {fraction: 0.5, youngs: 300}}
  particle: {parent: cell, per_parent: 100, radius: 0.02} # containment
fields:
  chemical: {frame: grid, res: 96, couples_to: cell}
operators:
  - {op: glide, at: cell, speed: 15.0, rot: 0.4}
  - {op: mls_mpm_mechanics, at: particle, n_grid: 128, substeps: 8, a_max: 800, drag: 4}
  - {op: deposit, at: "cell[type=soft]", to: chemical, rate: 1.0} # only pop 1 makes it
  - {op: diffuse, at: chemical, rate: 0.1} # field self-dynamics (was chemical.
    diffuse)
  - {op: decay, at: chemical, rate: 0.05}
  - {op: chemotax, at: "cell[type=soft]", from: chemical, gain: 150.0} # only pop 1 climbs it
schedule: [aggregate, [glide], deposit, diffuse, decay, chemotax, mls_mpm_mechanics]
```

The validator (prototype: `scenario_schema.py`) guarantees that a file which loads is runnable: every `op` exists; every `at/to/from` references a declared set/field; type `fractions` sum to one; every schedule token resolves. (One trap: never use a YAML 1.1 boolean key such as `on/off` — we use `at`.)

8 Operator catalog

Operators are the system’s vocabulary; it grows monotonically and every past spec keeps working. The catalog, grouped by what each operator changes (Section 4). `EMIT` is the temporal-integration state a dynamics operator’s returned delta represents — `velocity` / `acceleration` integrated by the engine, `mpm_acceleration` routed to the MPM substep, or `---` for an operator that writes a field, a relation, or a `heading` and returns no set delta:

op	kind	EMIT	behaviour
<i>Dynamics — move a set's state (return a delta)</i>			
signal	lateral	velocity	passive connectome: $dv_i = (-v_i + b_i + \sum_j W_{ij}\phi(v_j) + I_i)/\tau_i$
synapse_ode	lateral	velocity	edge-set synaptic state $ds_e/dt = h(s_e, v_{pre}, v_{post})$ (gathers endpoints along ι)
synaptic_current	aggregate	—	sum synapse output onto the postsynaptic neuron (Aggregate along ι_{post})
neuron_voltage	lateral	velocity	neuron ODE $\tau_i dv_i/dt = \text{intrinsic}_i + I_i^{\text{syn}} + I_i^{\text{ext}}$
attraction_repulsion	lateral	velocity	per-type difference-of-Gaussians interaction law
squared_law	lateral	acceleration	inverse-square law: Coulomb electrostatics <i>or</i> Newtonian gravity (law: charge/mass)
attractor_flow	lateral	velocity	ride a set along a strange-attractor field $\dot{x} = f(x)$ (system: lorenz/rossler/chen/...; 3D-only, no interaction)
cohesion, velocity_align, separation	lateral	acceleration	the three boids rules; their deltas sum to flocking
velocity_cruise	lateral	acceleration	Vicsek self-propulsion toward cruising speed v_0
drag	lateral	accel. mpm_accel.	viscous friction $-\kappa v$ (emit: routes to the engine or the MPM substep)
glide	lateral	velocity	active self-propulsion along the heading
gravity, mpm_anchor, mpm_spin	lateral	mpm_accel.	uniform / rest-anchor / solid-spin body forces (MPM substep)
bounce	lateral	—	specular wall/obstacle re-heading (writes heading)
chemotax	exchange	vel. mpm_accel.	climb/flee a field's gradient (emit: routes)
sense	exchange	—	Physarum 3-sensor pheromone steering (writes heading)
deposit	exchange	—	secrete/deposit set state into a field
polarity_align, polarity_flow_align	exchange	—	align heading to neighbours / to solved MPM flow
active_force	exchange	mpm_accel.	activation-gradient body force (cardiac contraction)
active_stress	exchange	—	activation $\rightarrow$ active-stress tensor (H.active_stress)
apply_material_map	exchange	—	image field $\rightarrow$ per-particle material (youngs)
agent_scatter, agent_gather, agent_remodel	exchange	— / vel.	two-way agent $\leftrightarrow$ material coupling
aggregate	aggregate	—	parent state = mass-weighted mean of children ($\uparrow$)
broadcast	broadcast	velocity	lift parent state onto its children ($\downarrow$)
mpm_strain, mpm_scatter, mpm_grid_update, mpm_gather	lat./exch./field	—	the decomposed MLS-MPM cycle (strain $\rightarrow$ scatter $\rightarrow$ grid solve $\rightarrow$ gather)
mls_mpm_mechanics	exchange	—	the same cycle as one fenced <i>transitional</i> or-acle (Section G)
<i>Fields — a field's own dynamics (kind=field; write the grid, return {})</i>			
diffuse, decay	field	—	Laplacian diffusion / linear evaporation
pacemaker	field	—	periodic scalar clock $p(t) \rightarrow \mathbf{H.signals}$
activation_pulse	field	—	clocked activation field: shared-clock $\times$ profile, or per-pixel delayed wave
playback	field	—	advance a prescribed (video) field one frame
<i>Relations — change the edge set E (kind=rewire)</i>			
radius_graph	rewire	—	neighbour graph: live pairs within a cutoff, rebuild each tick
<i>Entities — change the node set S (kind=structural)</i>			
cell_divide	structural	—	proliferation: wake a dormant buffer slot (occ) beside the mother

9 Natural language $\rightarrow$ simulation: the repeatable mapping

To make the LLM's compilation *repeatable* (same request $\rightarrow$ same structure), it follows fixed rules; the most useful:

1. “ N cells/agents” $\rightarrow$ a set with **n:** N .

2. “made of K X per cell” → a contained set with `parent/per_parent` (two levels).
3. “two types differing by P ” → `types:` with P as a per-type property.
4. “mechanical/elastic/rigid” → particles + the MPM operators; map stiffness to `youngs`.
5. “moves by boids / flock” → `cohesion+velocity_align+separation`; “wanders / motile” → `glide`.
6. “creates/secretates a chemical” → a `field` + `deposit`; “follows a gradient / chemotaxis” → `chemotax`; “ant / pheromone trails” → `sense` (Physarum 3-sensor).
7. “particles attract/repel, interact by type” → a `radius_graph` (the relation) followed by a lateral interaction op (e.g. `attraction_repulsion`); the per-type law parameters go in `types:`, read by the operator.
8. “**only population X does Y**” → the selector `at: cell[type=X]`. This is the highest-leverage primitive: it scopes behaviour to a subpopulation and lets two behaviours coexist.

Throughout, the mapping respects the set/field/operator split of Section 3: a *behaviour* becomes an operator, a *kind of node* a set (or a type within one), the *space* the `general:` block — never the reverse.

Part III — Building the repository

10 Repository contract

Plexus is not a collection of simulations. It is an interpreter for a typed intermediate representation. The repository must therefore be built around one rule:

New biology enters Plexus by adding registered entities, fields, or operators; never by adding simulation-specific logic to the engine.

The engine is the kernel. The spec is the program. The registry is the vocabulary. The validator is the contract gatekeeper. Operators are the only place where new dynamics live.

Every coding change must preserve this separation.

`spec.yaml` → `schema` → `registry` → `engine` → `trajectory`.

If a change requires the engine to branch on a simulation name, biological case, operator name, or field name, the abstraction has failed. The correct fix is almost always a new registered operator, a new entity schema, a new field class, or a stronger validator rule.

11 Layering

The repository is divided into strict layers.

- `models/base.py`: the contract layer: `Level`, `Field`, `Hierarchy`, `Operator`, and the semantic operator kinds.

- `models/registry.py`: the only name-to-code map. It registers entities, fields, and operators. The engine must resolve names only through this registry.
- `models/entities.py`: declares state schemas and rendering hints for entity kinds. It is the only source of truth for what columns of `Level.state` mean.
- `operators/*.py`: one operator per file, one concern per operator. Operators declare their `KIND`, `EMIT`, `REQUIRES_PARAMS`, `REQUIRES_TYPE_PROPS`, and optional mechanism-search meta-data.
- `fields/*.py`: field state classes. A field stores continuum state and geometry. Field dynamics are operators, not field methods.
- `schema.py`: validates the spec before execution. A spec that passes schema validation should be runnable by the engine.
- `engine.py`: the generic interpreter. It builds the hierarchy, instantiates registered operators, executes the schedule, sums deltas, and integrates each set once per tick.
- `plot.py` or `viz.py`: generic visualization over recorded sets and fields. It must never branch on simulation names.
- `archive.py` and `simulations/`: reproducible gallery and ledger: spec, seed, metrics, thumbnails, mechanism captions, and failure notes.

No layer may reach upward. The engine may import the registry; the schema may import the registry; operators may import base classes and geometry utilities. Operators may not import the engine. The schema may not execute simulations.

12 Operator kinds

Every operator must belong to exactly one kind. The kind is determined by what the operator changes.

Kind	Changes	Returns	Examples
<code>lateral</code>	set state within one set	<code>delta</code>	attraction, boids, drag
<code>aggregate</code>	parent state from children	<code>delta</code> or assignment	particle-to-cell centroid
<code>broadcast</code>	child state from parent	<code>delta</code>	cell force to particles
<code>exchange</code>	set-field coupling	<code>delta</code> or field write	deposit, sense, chemotaxis
<code>field</code>	field self-state	<code>{}</code>	diffuse, decay, playback, reaction
<code>rewire</code>	relations E	<code>{}</code>	radius graph, membrane graph
<code>structural</code>	entity membership $ S $	<code>{}</code>	divide, die

This table is normative. Do not reinterpret it locally.

Forbidden shortcuts.

- Do not implement field self-dynamics as `exchange`. Use `kind="field"`.
- Do not implement neighbour rebuilding inside a force operator. Use `rewire`.
- Do not implement division or death by resizing tensors. Use fixed buffers and `occ`.

- Do not implement friction as an integrator option. Use a **drag** operator.
- Do not implement biological behaviour in the engine. Use an operator.

13 The integration invariant

The engine is the only code allowed to integrate a set’s state — position and velocity for a spatial set, or a schema-declared scalar such as voltage or concentration for a non-spatial one. Which block is integrated is read from the state schema, not hardcoded to **pos/vel**; the world boundary is applied only to spatial blocks.

A dynamics operator never writes a set’s integrated state (**pos/vel**, or a scalar like **v**) directly. It returns a delta:

$$\Delta_i = \begin{cases} \text{a velocity} & \text{if EMIT} = \text{velocity}, \\ \text{an acceleration} & \text{if EMIT} = \text{acceleration}, \\ \text{a substep body acceleration} & \text{if EMIT} = \text{mpm.acceleration}. \end{cases}$$

The engine sums the deltas for a set and integrates once at the end of the tick; **mpm.acceleration** deltas are read instead by the MPM substep (as the external acceleration a_{ext}) and are never engine-integrated. This is one vocabulary, shared by the class attribute **EMIT** and the spec **emit**: override with no translation table; **EMIT** = **None** (a **field** / **rewire** / **structural** operator, or an exchange that writes a field) emits no set delta at all.

Allowed in-place mutations are limited to:

- relations: **edge_index** by **rewire** operators;
- fields: **grid** by **field** or set-to-field exchange operators;
- membership: **occ**, **birth**, **lineage**, and per-node buffers by **structural** operators;
- auxiliary control state: e.g. **heading**, provided it is not part of the engine-integrated state.

Everything else must be returned as a delta.

Engine guard. The engine must check this invariant around every non-mutating operator. The guard must not be disabled globally just because one structural operator exists. Instead, only operators that explicitly declare permission to mutate integrated state may do so. The preferred rule is

```
MAY_MUTATE_INTEGRATED_STATE = False
```

by default. Structural operators that truly need to rewrite state must opt in explicitly and document why.

14 Selector and occupancy rule

Every operator acting on a set receives a selector mask. Every operator must also respect occupancy. The effective mask is always

$$m_i = \mathbf{1}_{\text{selector}}(i) w_i, \quad w_i = \text{occ}_i.$$

This applies to all operators, including operators that mutate auxiliary state such as `heading`. Dormant nodes must not move, sense, deposit, attract, repel, divide, die again, or contribute to aggregates. Required pattern:

```
m = selector_mask * lvl.occ
```

For message passing, dormant senders and receivers must both be handled: sender occupancy weights messages; receiver occupancy masks the output.

15 Field rule

A field is pure state: a grid, geometry, and possibly external data. A field does not own its dynamics.

- `deposit`: `set` $\rightarrow$ `field`, `kind="exchange"`.
- `sense` or `chemotax`: `field` $\rightarrow$ `set`, `kind="exchange"`.
- `diffuse`, `decay`, `playback`, `react`: `field` $\rightarrow$ `field`, `kind="field"`.

Retired schedule forms such as `chemical.diffuse` are forbidden. Field updates are ordinary registered operators:

```
operators:
- {op: diffuse, at: chemical, rate: 0.2}
- {op: decay, at: chemical, rate: 0.01}
schedule: [deposit, diffuse, decay, sense]
```

The schema and engine must agree. If the schema accepts a schedule token, the engine must run it. If the engine rejects a token, the schema must reject it first.

16 Aggregate and broadcast rule

Aggregate and Broadcast are not optional conveniences. They are primitive operators of the calculus.

Aggregate. An aggregate operator moves information upward along a parent map, $S_\alpha \rightarrow S_\beta$. It must use `Level.parent` and occupancy-weighted reductions, and must not assume a linear scale ladder. Required form:

$$x_\beta(b) = \frac{\sum_{i:\pi(i)=b} w_i x_i}{\sum_{i:\pi(i)=b} w_i}.$$

Broadcast. A broadcast operator moves information downward along the same parent map, $S_\beta \rightarrow S_\alpha$. It must use `Level.parent` to index parent state for each child, and must respect child occupancy.

Aggregate and Broadcast are implemented as registered operators, not as special string builtins in the engine. The schedule names them like any other operator.

Along any named map. The reduction and lift are defined by the *map*, not by any one role. An operator may run them along a *pre/post* map (`Level.pre`, `Level.post`) instead of the parent map (`Level.parent`) — both are index buffers of the same shape, so the reduction primitive is identical. This is how a stateful connection signals: `synaptic_current` is an Aggregate along ι_{post} , $I_i = \sum_{e: \iota_{\text{post}}(e)=i} g(s_e, v_i)$, and the endpoint voltages a `synapse_ode` needs are a Broadcast/gather along ι_{pre} and ι_{post} . Occupancy weighting is unchanged: dormant edges and dormant endpoints contribute nothing.

17 Structural rule: divide and die

Structural operators change membership, not tensor shape. All sets are allocated at a fixed buffer size; a node exists when `occi > 0`.

Divide. Division activates dormant slots. It must use `Level.spawn` or an equivalent central primitive that copies all per-node buffers safely. A division operator must not hand-copy only `state`; it must preserve all relevant per-node buffers: type, parent, mass, deformation state, lineage, and domain-specific buffers.

Die. Death sets `occ` $\rightarrow$ 0. If a level has mass, death also sets mass to zero. A dead node should no longer contribute to rewire, lateral interactions, exchange, aggregation, visualization, or metrics.

No resizing. Structural operators never call tensor concatenation to change the number of nodes. They wake or retire fixed slots.

18 Boundary and geometry rule

The world owns geometry.

- Domain width, height convention, boundary type, and obstacles live in `general::`.
- Operators read `H.world_width`, `H.periodic`, and `H.obstacles`.
- Boundary conventions must be shared by all spatial operators.
- A world is $[0, W] \times [0, 1]$ unless the base geometry contract is changed globally.

No operator may define its own incompatible notion of the domain. For finite-difference field gradients, periodic worlds may wrap; wall worlds must not wrap — use boundary-aware padding.

19 State schema rule

Operators must not hard-code state columns unless they are explicitly limited to an entity whose schema is fixed. Preferred form:

```
pos = lvl.get("pos")
vel = lvl.get("vel")
```

The meaning of state columns belongs to `@register_entity`, not to an operator; a new entity kind declares its schema once. Forbidden pattern (except in temporary code with a comment explaining why the entity schema is not yet available):

```
pos = lvl.state[:, :2]
vel = lvl.state[:, 2:4]
```

20 Type and parameter rule

A type is a parametric variant inside one set; a sort is a different set.

- Different state space or different admissible operators: new set.
- A connection with its own dynamical state is a new set too — an *edge-set* — joined to its endpoints by **pre/post** maps rather than a single parent map. A *passive* connection stays a relation (**edge_index**+attribute), not a set.
- Same state space and same operators, different parameter values: type within one set.
- Per-type properties belong in **types:**.
- Operators read per-type properties only through declared **REQUIRES_TYPE_PROPS**.

When assigning type fractions to a buffer, rounding must not silently leave unassigned nodes. The final type receives all remaining nodes.

21 Schema–engine consistency

The schema is the contract gatekeeper; the engine must not discover avoidable errors at runtime. The schema must reject:

- unregistered operators; unknown sets or fields; unknown operator kinds;
- missing **REQUIRES_PARAMS**; missing **REQUIRES_TYPE_PROPS**;
- schedule tokens that do not resolve; retired builtins;
- conflicting prediction orders on the same set; invalid type fractions;
- field operators whose **at:** is not a field; set operators whose **at:** is not a set.

A validated spec should not hit **NotImplementedError** in the engine except for explicitly marked experimental features.

22 Mechanism metadata rule

Every operator should be captionable; this supports the mechanism-search ledger. Operators may declare **MECHANISM_TAGS** and **PARAM_ROLES**. These are not used by the engine — they are used by documentation, the simulation ledger, and agentic mechanism search:

```
MECHANISM_TAGS = ["long_range_attraction", "short_range_repulsion", "coarsening"]
PARAM_ROLES = {"sigma": "interaction_length"}
```

The goal is to make every spec translatable into mechanistic language. (An earlier **MORPHOLOGY_PRIOR** field — a fixed per-operator guess at emergent shape — was removed: morphology is set by the

parameters of a spec, not by the operator, so the same operator yields a bound cluster, a featureless disc, or a spiral disc depending on its couplings.)

23 Operator naming rule

An operator’s name states its *mechanism*, never its *wiring*. Which entities an operator connects is already declared per invocation — **from:** and **to:** name the source and target set or field, and **EMIT** names the state its delta updates — so encoding the endpoints in the name (**agent_to_mpm**, **pulse_to_active_stress**, **field_to_agent**) is redundant and does not scale: every new pair of endpoints would mint a new name (**particle_to_grid**, **grid_to_mesh**, **field_to_particle**, ...). Name the algorithm and let the spec route it:

```
- op: scatter          # mechanism: particle -> grid transfer
  from: cell
  to: mpm_particle
- op: gather           # the same transfer, wired the other way
  from: mpm_particle
  to: cell
```

Thus the MLS-MPM transport pair **p2g/g2p** becomes **mpm_scatter/mpm_gather**, and the agent↔grid pair **agent_to_mpm/mpm_to_agent** becomes **agent_scatter/agent_gather** (subsystem-qualified, since their scatter/gather kernels differ from the MLS-MPM ones — same mechanism family, different contract, so kept separate rather than merged), while **pulse_to_active_stress/pulse_to_contraction** are **active_stress/active_force** — two constitutive laws distinguished by the quantity they emit (a stress tensor $-A\hat{n}\hat{n}^\top$ vs. a body force), not by their input: the activation source may be calcium, voltage, or a morphogen, so the name must not hard-code **pulse**. The families **scatter/gather/deposit/sample/aggregate/broadcast** are mechanisms; the graph already knows who talks to whom.

Naming rule. Operator names describe mechanisms; **from:** and **to:** describe routing. A directional **X.to_Y** name is admissible only when $X \rightarrow Y$ is itself the scientific mechanism, not merely its endpoints.

The test: if two operators differ only in their endpoints and share a mechanism, they are *one* operator wired two ways, not two names — exactly the merge-by-contract rule of §27 read through the routing lens. The name answers only “what does this compute?”; **from:/to:/EMIT** answer “between what, in which direction.”

24 Build loop for a coding agent

A coding agent must follow this loop for every new capability.

1. Identify the category of the requested feature: state, field, relation, entity membership, hierarchy, world, visualization, or validation.
2. Choose the correct home: operator, field, entity schema, validator, engine, or plotter.
3. Before editing code, write the expected spec line.
4. Implement the smallest registered primitive that makes that spec line valid.
5. Add **REQUIRES_PARAMS**, **REQUIRES_TYPE_PROPS**, and mechanism metadata.

6. Add schema validation if the primitive introduces a new contract.
7. Run a minimal spec.
8. Verify three things: the spec validates, the engine runs, and the intended morphology or invariant is measured.
9. Add the run to the gallery with spec, seed, metrics, thumbnail, and mechanistic caption.
10. Do not edit the engine unless the new feature is impossible to express as a registered primitive.

25 Common failure modes and corrections

Failure	Correction
Operator directly updates <code>pos</code>	Return a delta; let the engine integrate.
Field self-update registered as exchange	Use <code>FieldUpdate</code> , <code>kind="field"</code> .
Dormant nodes still act	Multiply the selector mask by <code>occ</code> .
Retired builtin accepted by schema	Remove it from schema and engine; use a registered operator.
Rewire hidden inside a force law	Split into <code>radius_graph</code> + a lateral operator.
Structural op resizes tensors	Use a fixed buffer + <code>occ</code> .
One structural op disables all guards	Guard per operator, not globally.
Operator hard-codes <code>state[:, :2]</code>	Use <code>lvl.get("pos")</code> and entity schemas.
Type fractions leave leftovers	Assign the remainder to the final type.
Engine branches on a simulation name	Move the behaviour into a registered operator.
Plotter branches on a simulation name	Render generically from the data layout and render hints.
Metric says the run succeeded but the morphology is wrong	Add intent-level verification.

26 Definition of done

A new operator or capability is complete only when all of the following are true:

- It is registered. Its kind is correct. Its capability contract is declared.
- It respects selector masks and occupancy.
- It does not mutate integrated state unless explicitly structural.
- It has a minimal spec. The spec validates. The engine runs without special cases.
- The output has an intent metric.
- The result is archived with spec, seed, metrics, and caption.

If any item is missing, the feature is not part of Plexus yet — it is only prototype code.

27 From prototypes to the Plexus core

Plexus is developed through scientific prototypes: embryo, cardio, active matter, and other domain-specific campaigns routinely introduce operators before their final abstraction is known. A prototype clones an existing model — a vertex tissue, a reaction–diffusion patterner, a self-propelled-Voronoi sheet — reproduces it as a throwaway operator set, and explores it freely. Code is therefore *not* moved into the core merely because it works in one prototype. Moving code from a prototype into Plexus is not copying useful files; it is **promoting a mechanism into a stable operator language**. Promotion happens only after the mechanism’s family, contract, naming, dimensional assumptions, and tests have stabilized. The rule is simple: *prototype freely, promote conservatively*.

Promotion passes through five gates.

1. **Promote families, not isolated operators.** Before a mechanism is moved, it is assigned to a family — a closed taxonomy declared in the registry (`motion`, `interaction`, `kinetics`, `polarity`, `fields`, `mechanics`, `mpm`, `coupling`, `hierarchy`, `growth`, `topology`). The family is a keyword on `@register_operator(..., family=...)`, and `OPERATOR_FAMILIES` is the closed set an operator’s family must belong to. An operator enters the core only when its family is clear; this keeps the registry a browsable set of reusable physical primitives (`operators_by_family("mechanics")`) rather than a flat pile of one-off biological events. Crucially a family names a *mathematical* primitive — message passing along a map (`interaction`), a local state ODE (`kinetics`), a reduction along a map (`hierarchy`) — never a biological interpretation: neuronal signalling, metabolism, and gene regulation are all the same `kinetics-plus-interaction` pattern, so there is deliberately no `signalling` family. Biology lives in the *levels* and *specs*, not the family taxonomy. A single check, `tools/audit_operator_registry.py`, enforces the whole contract at once — every operator has a valid family and explicitly declares the five class attributes (§23), `EMIT` is one of the canonical values, `SUPPORTED_DIMS` is a subset of $\{2, 3\}$, no deprecated alias appears in a source spec, and every `config/` spec still resolves — so the vocabulary cannot silently drift as the registry grows.
2. **Merge by contract, not by surface similarity.** Two prototype operators collapse into one canonical operator with explicit options only when they share the same contract and differ solely in routing, source field, or mode — for example `chemotaxis+chemo_force→chemotax`, `mpm_drag+drag→drag` with an `emit` option, and `pulse_stimulus+phase_delay_pulse→activation_pulse`. When the contract differs they stay separate and share only backend utilities: `active_force` emits an `mpm_acceleration`, whereas `active_stress` writes an active stress and emits nothing — those must not become one muddy mode-dependent operator. Merging is thus governed by physical and computational semantics, never by superficial code similarity.
3. **Stabilize names before promotion.** Every promoted operator receives one canonical name and one canonical vocabulary: its Python class, registry key, YAML name, emitted prediction type, state fields, schema path, scorecard metric names, and the terminology in this manuscript must all agree. The `EMIT` refactor is the model — `velocity`, `acceleration`, `mpm_acceleration`, with no synonyms, no legacy aliases, and no `first_derivative/second_derivative` drift. Legacy synonyms are removed *at* promotion time rather than preserved indefinitely, so that Plexus stays a language and not a collection of historical aliases.
4. **Require 2D → 3D design at promotion time.** The 3D implementation need not exist immediately, but the interface must make 3D possible. Each promoted operator declares `SUPPORTED_DIMS` ($[2]$ or $[2, 3]$) and records which quantities are scalars, vectors, and tensors,

which assumptions are 2D-only, and what the 3D generalization would be. A planar angle θ or a signed chirality is 2D-specific and must state its axis/vector/handedness generalization; `Field.sample` and `Field.grad_at` are naturally dimension-generic. The goal is to prevent 2D geometry from silently entering the core abstraction.

5. **Promotion requires tests and paper reconciliation.** A promotion is complete only when code, specifications, tests, and documentation move together. Behaviour-preserving migrations require a byte-identical trajectory check against the originating prototype on a representative spec; new mechanisms require a no-op/ablation test, schema validation, and at least one minimal live spec. The operator catalog and this manuscript are then regenerated or reconciled so the paper describes the live registry rather than an older prototype.
6. **Cite the source paper and repository.** Because a prototype begins by cloning a published model, the promoted operator must carry that provenance: the originating paper and code repository are recorded in the operator’s docstring and in this manuscript, so a reader can trace each core primitive back to the science it was built from. Promotion inherits credit, it does not erase it.

As a worked example, the inverse-square `squared_law` operator is a single such promotion: it unifies the electrostatic Coulomb law (ported from ParticleGraph’s `PDE_E`) with softened Newtonian gravity (from Philip Mocz, *Create Your Own N-body Simulation (With Python)*, 2020, vendored under `papers/nbody-python/`) into one `law`-switched operator. Both are the same contract — an add-aggregated pairwise $1/r^2$ force emitted as an acceleration — differing only in the source charge (`charge` vs. `mass`), the sign/receiver convention (like-repel vs. attract), and the summation range (neighbour graph vs. all-pairs). The Coulomb path is byte-identical to the pre-merge operator; the gravity path reproduces the prototype’s force exactly. The galaxy prototype is thereby retired into the core as a spec, not a fork.

The `attractor_flow` operator is a second such promotion, out of the `prototype/attractors` campaign: ten classic strange attractors (Lorenz, Rössler, Aizawa, Sprott B, Halvorsen, Thomas, Chen, Chua, Dadras, Rabinovich–Fabrikant) collapse into *one* `system`-switched operator by the merge-by-contract rule — they share the same contract, a per-point autonomous field $\dot{x} = f(x)$ emitted as a `velocity`, differing only in the field’s closed form and constants. It is filed under `motion` (a prescribed kinematic velocity rule, like `drag/glide`), declares `SUPPORTED_DIMS= [3]` on principle rather than as an unfinished interface (a continuous planar flow cannot be chaotic, by Poincaré–Bendixson), and carries each system’s originating paper in its docstring. The ten prototype specs retire into `config/attractors/` and render through the existing generic 3D splat, no new plotter.

This promotion discipline is the same reconvergence process the repository was built for: sibling prototype frameworks that had drifted apart are pulled back into one shared vocabulary, one operator at a time. Prototypes are where mechanisms are discovered; the core is where they become a language.

28 The principle

Plexus grows by expanding its vocabulary, not by weakening its grammar.

A coding agent should therefore prefer adding one small, typed, registered primitive over adding a clever shortcut. Shortcuts make one simulation work. Primitives make the language larger.

Appendix

A Glossary

Concept	Set/operator term	Symbol	Code
collection of like nodes	set (a level)	S_k	<code>Level</code>
a single node	element	$x \in S_k$	row of <code>Level.state</code>
what each element carries	state	named blocks of S_k	<code>Level.state</code> + schema
learnable per-node code	embedding	a_i	<code>Level.embedding</code>
map between sets	named map	$m: S_a \rightarrow S_b$	index buffer (<code>.parent</code> , <code>.pre...</code>)
all maps of a container	map collection	$M = \{m_1, m_2, \dots\}$	—
containment (map role)	parent map	$\pi_k: S_k \rightarrow S_{k+1}$	<code>.parent</code> , <code>.parent_name</code>
incidence (map role)	incidence map	$\iota: S_e \rightarrow S_k$	<code>.pre</code> , <code>.post</code>
connection with state	edge-set	S_e	<code>Level (edge_set)</code>
within-set links	relation	$E \subseteq S_k \times S_k$	<code>Level.edge_index</code>
continuum quantity	field	$f: \Omega \rightarrow \mathbb{R}^c$	<code>Field</code>
dynamics (GNN)	operator: ODE+Laplacian	$\mathcal{O}$	<code>Operator</code>
operator interface	contract	state $\rightarrow$ state	the op name
choice of dynamics for a contract	implementation	—	<code>implementation:</code>
a simulator implementing contracts	backend	—	(Jaxley, NEURON)
within-set op	lateral	$\mathcal{O}_E$	<code>Lateral</code>
up op	reduction over children	$\sum_\pi$	<code>Aggregate</code>
down op	lift	π^*	<code>Broadcast</code>
set $\leftrightarrow$ field op	transfer	$\mathcal{S}, \mathcal{G}$	<code>Exchange</code>
rebuild the relation	relations: rewire	$\mathcal{R}: \rightarrow E$	<code>Rewire</code>
make an element	entities: division	$\mathcal{B}: x \mapsto \{x', x''\}$	<code>Divide</code>
remove an element	entities: death	$\mathcal{D}: x \mapsto \emptyset$	<code>Die</code>
per-element occupancy	occupancy	$w_i \in [0, 1]$	<code>Level.occ</code>
the container	sets, fields, maps	$(S_\bullet, F_\bullet, M_\bullet)$	<code>Hierarchy</code>
the model	composition of ops	$\mathcal{O}_n \circ \dots \circ \mathcal{O}_1$	<code>Schedule</code>

B Plexus and Lean

At first glance Plexus and a proof assistant such as LEAN appear unrelated. One simulates living systems; the other verifies mathematics. Yet they share a surprisingly similar architecture. Both seek to construct complex objects from a small set of primitive building blocks while preserving correctness. Understanding this correspondence helps clarify what Plexus actually is: not merely a simulator, but a formal system for describing biological models.

B.1 A brief introduction to Lean

Lean is an interactive theorem prover developed to formalize mathematics. A user states a theorem and constructs a proof. Unlike a traditional mathematical proof written for humans, a Lean proof must be expressed in a precise formal language that the Lean kernel can verify mechanically.

For example, the mathematical statement

$$a + b = b + a$$

can be expressed as a theorem in Lean:

```
theorem add_comm (a b : Nat) : a + b = b + a := ...
```

The dots represent a proof. Once supplied, Lean checks every step and certifies that the conclusion follows from the rules of the system.

Lean is built around several layers.

1. **Types.** Every object belongs to a type: natural numbers, real numbers, sets, functions, and so on.
2. **Terms.** Concrete objects inhabit types. The number 3 is a term of type `Nat`. A function is a term of a function type.
3. **Theorems.** A theorem is a proposition that one wishes to prove.
4. **Proof terms.** A proof is itself an object. Under the Curry–Howard correspondence, propositions behave like types and proofs behave like terms inhabiting those types.
5. **The kernel.** A small trusted program checks that a proof is valid.
6. **Tactics.** Convenient automation that helps construct proofs. Tactics can be sophisticated and even unreliable internally because the kernel checks the result afterward.
7. **Mathlib.** A large library of previously formalized mathematics.

The key idea is separation of concerns. The user writes proofs, tactics help construct them, and the kernel verifies them. Trust resides in the kernel rather than in the process that produced the proof.

B.2 The structural correspondence

Plexus exhibits a remarkably similar decomposition. The correspondence is not perfect, but it is useful.

Lean	Plexus	Role
Types	Sets S_α , their state, Fields F , and maps M	primitive objects
Terms	Entities (cells, particles, metabolites)	instances of those objects
Functions	Operators	transformations
Proof composition	Schedule	composition of transformations
Type checker	Schema validator	correctness gate
Tactics	LLM compiler	search / generation layer
Theorem	Intent	desired outcome
Proof term	Simulation specification	executable description
Kernel	Engine	trusted interpreter
Mathlib	Operator registry	reusable library of primitives

Several rows deserve explanation.

A set in Plexus plays a role similar to a type in Lean. Just as Lean distinguishes natural numbers from matrices or groups, Plexus distinguishes cells, particles, metabolites, nuclei, and fields.

Operators declare which kinds of objects they act upon, exactly as functions declare their input types.

Entities are analogous to terms. A particular cell is an element of the set `cell`, just as the number 3 is an element of the type `Nat`.

The operator registry resembles a mathematical library. New functionality is added by registering a new operator rather than modifying the engine itself. Existing simulations continue to work because the vocabulary only grows.

The strongest correspondence is architectural. The simulation specification (`spec.yaml`) is generated by a human or by an LLM. It is not trusted. The schema validator checks that the specification is well formed before execution. Only after passing validation is it handed to the engine. This mirrors the relationship between tactics and the Lean kernel.

B.3 Why the analogy is imperfect

Despite these similarities, Plexus is not a theorem prover.

Lean is fundamentally concerned with *truth*. Given a proposition P , Lean determines whether a proof exists. If the kernel accepts the proof, the theorem is established with mathematical certainty.

Plexus is concerned with *dynamics*. Given an initial state and a collection of operators, Plexus computes

$$x_{t+1} = \Phi(x_t).$$

The result is not a proof but a trajectory.

This difference has important consequences.

First, the Plexus validator checks only *well-formedness*, not correctness. A specification that passes validation may still represent a poor biological model. The analogue of theorem proving in Plexus is not validation but experimental verification.

Second, Lean computations terminate. A proof either checks or it does not. Plexus simulations may be unstable, chaotic, or sensitive to parameters. The engine integrates a dynamical system rather than reducing an expression to normal form.

Third, Plexus is differentiable. Gradients flow through operators, schedules, and rollouts. This allows optimization of model parameters and even biological designs. There is no direct analogue of differentiation in theorem proving.

B.4 What Plexus borrows from theorem provers

The value of the comparison is therefore not that Plexus proves theorems. Rather, it inherits a design philosophy from systems such as Lean.

The first lesson is to separate a symbolic description from its execution. A simulation should be represented explicitly rather than being scattered across code.

The second lesson is to separate a trusted kernel from untrusted generation. The LLM may produce a simulation specification, but correctness of the specification is established by the validator and the engine, not by the model that wrote it.

The third lesson is that a small set of primitives can support a large space of constructions. Lean builds mathematics from types, functions, and proof rules. Plexus builds biological models from sets, state, fields, maps, operators, and schedules.

B.5 A language and calculus of living systems

Seen from this perspective, Plexus is best understood as a formal language and calculus for living systems.

The language is the simulation specification. It provides a symbolic vocabulary for describing sets, their state, fields, maps, and schedules.

The calculus is the collection of operators acting on those objects. Lateral, Aggregate, Broadcast, Exchange, Field, Rewire, Structural (Divide and Die), and Seed are the primitive transformations from which larger biological mechanisms are assembled.

The engine plays the role of an interpreter, turning symbolic descriptions into dynamical trajectories.

This is ultimately the deepest connection to Lean. Both systems seek a compact set of primitives from which a much larger space of structures can be constructed. Lean does this for mathematics. Plexus attempts to do it for biological systems.

C Plexus, World Models, and Mechanism Search

The goal of Plexus is not merely to simulate biological systems. It is to provide a language in which biological mechanisms can be expressed, compared, and discovered.

This distinction became apparent during an agentic modeling experiment on *Dictyostelium* aggregation. An agent was tasked with reproducing a real biological morphology from experimental observations. The search successfully explored many parameterizations of chemotaxis, adhesion, relay signaling, saturated sensing, density repulsion, and deformable mechanics. Several mechanisms were falsified, and one family of models transiently reproduced the target morphology before collapsing at longer times.

At first sight this appeared to be a failure of optimization. In retrospect it revealed a more important limitation: the search explored parameters within a mechanism, but not the space of mechanisms themselves.

Human observers immediately proposed alternative explanations:

- long-range attraction between aggregates,
- external radial fields,
- soft-body fusion,
- coarsening dynamics,
- three-dimensional aggregation projected into two dimensions.

None of these hypotheses were explored. The reason was not that they were unlikely, but that they were absent from the search space. The loop performed local refinement around an existing model rather than proposing qualitatively different explanatory worlds.

The lesson is that scientific discovery requires two distinct searches:

$$\text{mechanism search} \gg \text{parameter search}.$$

The first determines *what kind of world* might explain an observation. The second determines *which parameters* within that world are most consistent with the data.

D World Models and Plexus

Modern world models approach this problem from the opposite direction.

A predictive model such as JEPA or a video foundation model learns a latent representation

$$\text{video} \longrightarrow z_t \longrightarrow z_{t+1},$$

where the latent state z_t is optimized for prediction. Such models can be extremely successful at forecasting future observations, but the latent representation is not required to correspond to interpretable biological mechanisms.

Plexus instead begins from explicit objects:

$$(S_{\bullet}, F_{\bullet}, M_{\bullet})$$

— sets, fields, and an open collection of named maps (each playing a containment or incidence role)
 — and explicit operators acting on them. Cells, particles, fields, and interactions remain visible throughout the computation.

The difference is therefore not prediction versus simulation. Both can predict.

The difference is the object being searched.

	World Models	Plexus
Primitive representation	latent variables	sets, state, fields, maps, operators
Search space	latent trajectories	mechanisms and parameters
Primary strength	prediction	hypothesis testing
Interpretability	low–moderate	high
Counterfactuals	implicit	explicit
Scientific hypotheses	difficult to extract	native objects

The two approaches should therefore be viewed as complementary rather than competitive. A world model discovers regularities. Plexus evaluates candidate mechanisms capable of generating those regularities.

E A Mechanistic World Model

The Dicty experiment suggests that a successful discovery system requires an intermediate layer between observations and simulations.

Rather than learning latent dynamics directly, the objective is to learn a mapping

$$\text{simulation specification} \longrightarrow \text{mechanistic description}.$$

Every simulation in the Plexus gallery already contains three linked objects:

(specification, trajectory, mechanistic explanation).

For example:

“Long-range attraction produces coarsening and eventual fusion into a dominant aggregate.”

or

“A reaction–diffusion field generates filamentary transport networks through local activation and long-range inhibition.”

These descriptions form a growing knowledge base connecting operators and morphologies.

The objective is therefore not to learn

video $\rightarrow$ future video,

but rather

video $\rightarrow$ mechanistic hypotheses.

This representation remains interpretable because its primitives are the same objects used by the simulator itself.

F The Mechanism Search Tree

Once mechanisms are represented explicitly, the search process changes.

Traditional optimization explores a parameter space:

$\theta_1, \theta_2, \dots$

for a fixed model.

Plexus instead organizes discovery as a hierarchical search over mechanisms.

Observation $\rightarrow$ Mechanism Family $\rightarrow$ Mechanism Composition $\rightarrow$ Parameter Refinement.

The resulting search tree resembles

```
Explain observed morphology
|
+-- Long-range attraction
|   +-- attraction radius
|   +-- attraction strength
|
+-- Radial field
|   +-- field shape
|   +-- sensing gain
|
+-- Soft-body fusion
|   +-- Young's modulus
```

```

|   +-- surface tension
|
+-- Chemotactic relay
    +-- diffusion
    +-- decay

```

A bandit strategy such as UCB can then allocate computation not only between parameter values but also between competing explanatory worlds.

The crucial distinction is that branches correspond to scientific hypotheses, not merely numerical settings.

G Roadmap

The proposed roadmap consists of four stages.

Stage 1: Mechanistic Captioning. Construct a library of simulations spanning the operator vocabulary of Plexus. Each simulation is paired with a structured mechanistic description.

$$\text{spec} \leftrightarrow \text{mechanism} \leftrightarrow \text{video}.$$

Stage 2: Mechanistic Retrieval. Given a new observation, retrieve simulations with similar morphology and collect their associated explanations.

Stage 3: Mechanism Search. Use the retrieved explanations to initialize a mechanism search tree. The agent explores families, compositions, and parameterizations using simulation-guided evaluation.

Stage 4: Mechanistic World Models. Train models that map observations directly to distributions over candidate mechanisms. The world model no longer predicts future pixels. Instead, it predicts which explanations are worth testing.

$$\text{Observation} \rightarrow \text{Mechanism Prior} \rightarrow \text{Plexus Search} \rightarrow \text{Simulation} \rightarrow \text{Falsification}.$$

In this view, Plexus is neither a simulator nor a world model alone. It is a language in which candidate worlds can be written, searched, and falsified.

Construct	Semantics	Example
general: — the run and the world (the space + the clock)		
name	run identifier	name: aggregate
seed	RNG seed — fixes determinism	seed: 0
n_frames, dt	ticks to run; the time step	n_frames: 250
boundary	the space: periodic (torus) or wall (clamp)	boundary: periodic
world	domain width W ; world is $[0, W] \times [0, 1]$ ($W > 1$ longitudinal)	world: 4.0
obstacles	static walls (rects or $[cx, cy, r]$ circles); a BC for the MPM grid + field diffusion	[[.3, 0, .36, .7]]
<i>Set</i> (S_k — an entity level / ECS entity)		
n	size of a top-level set	n: 100
parent	the set that contains this one — a containment edge $\pi_{A \rightarrow B}$ (many sets may share a parent)	parent: cell
per_parent	children per parent	per_parent: 100
role	this set's role inside its parent (membrane / cytoplasm / nucleus...)	role: membrane
types	named sub-populations: fraction + per-type properties	{soft: {fraction: .5, youngs: 12}}
<i>Field</i> (a continuum)		
frame	discretization (a grid)	frame: grid
res	grid resolution	res: 96
diffusion, decay	PDE coefficients	diffusion: 0.1
couples_to	the one set this field exchanges with	couples_to: cell
source	fixed source region (imposes a gradient)	source: [.82, .82, 1, 1]
init	uniform initial fill (self-generated runs)	init: 1.0
navigation	geodesic (precomputed external gradient) or dynamic (self-generated)	navigation: geodesic
<i>Operator</i> (a system; one per list entry)		
op	which registered operator (contract) to run	op: sense
implementation	which <i>implementation</i> of that contract to run (default: the only one)	hodgkin_huxley
at	selector — the subset it acts on	at: cell[type=soft]
to/from	field an Exchange writes to / reads from	from: chemical
<params>	operator-specific arguments	gain: 150
<i>Selector</i> (sub-grammar)		
set	every node of the set	cell
set[attr=val]	nodes matching attr (re-checked each tick)	cell[loaded=1]
<i>Schedule</i> (sub-grammar; per-tick order)		
operator name	run that operator this tick	sense
[a, b]	a group run in the same stage	[glide, sense]
aggregate	a registered op: Aggregate $\uparrow$ (parent = mean of children)	aggregate
diffuse	a field op: advance that field one step (Laplacian)	diffuse
plotting: (render style only — never affects dynamics)		
colormap, point_size, background	how the sets are drawn	background: black

Table 2: The simulation language. Each block has its own home in the spec: **general:** is the world and clock, **sets:** are the entities, **operators:** the dynamics, **schedule:** the composition, **plotting:** the style. Integration is implicit (each set is integrated once at the end of a tick), so